Production UI patterns, from demo to mainnet.

WHAT SEPARATES A HACKATHON DEMO FROM AN APP USERS TRUST.

SESSION GOALS

Design a Solana frontend that survives real users, real congestion, and real RPC failures.

The shift nobody warns you about

DEMO REALITY

- One wallet, one network, one RPC, no users.
- "Happy path" dominates the screen.
- Errors are console.log'd.

PRODUCTION REALITY

- Congestion, rate limits, wallet rejects, dropped connections.
- Happy path is **the minority of sessions**.
- Errors are a first-class UI concern.

> If your UI only looks good during success, it isn't done.

A production UI tells the truth about every state — fast, slow, failed, expired.

State, by the numbers

8

STATES PER TX

2

RPC ENDPOINTS MINIMUM

0

INFINITE SPINNERS ALLOWED

States: **idle → preparing → signing → submitting → confirming → confirmed | expired | error**.

Every state gets a label, a color, and a next-step CTA.

RPC reliability drives UI design

FACTS FROM OFFICIAL DOCS

- Public mainnet RPC is **rate-limited** and carries **no SLA**.
- Private RPC + a fallback is the minimum for production.
- Congestion is real and **affects your UI**, not just your backend.

```
async function withRpcFallback<T>(
  run: (c: Connection) => Promise<T>,
) {
  try { return await run(primary); }
  catch { return await run(fallback); }
}
```


Source: Production readiness

Error taxonomy (the ones you must own)

CATEGORY	ORIGIN	UI RESPONSE
Wallet rejected	User	Soft notice, no guilt. "No changes. Try again when ready."
Wallet not installed	Browser	Direct link to install the wallet.
RPC rate-limited	Network	Retry via fallback. Show a "network is busy" banner.
Transaction expired	Timing	Dedicated expired state. Retry with fresh blockhash.
On-chain failure	Program	Show error details. Link to Explorer for transparency.

Priority fee UX, in one flow

- Expose **Off / Auto / Manual** as a segmented control.
- Default to **Auto** for beginners.
- Show a one-line explanation inline, not hidden in a tooltip.
- Display the final cost **before** the signature request.

> Priority fees are not optional UI. During congestion, they are the reason a transaction gets included at all.

Source: Priority fees

Accessibility is not optional

WCAG 2.1 AA BASICS

- Color contrast for all text and icons.
- Keyboard-operable inputs and buttons.
- Visible focus states.
- ARIA live regions for status updates.

WHY IT MATTERS ON SOLANA

- Money flows need to be usable for everyone.
- Screen readers must announce each transaction stage.
- Fast UIs without accessibility are still **inaccessible**.

Source: WCAG 2.1

Optimistic UI — carefully

OK FOR

- Toggling a "like", re-ordering a list, pre-selecting an option.
- Anything **reversible** without on-chain cost.

NEVER OK FOR

- Money movement.
- NFT minting.
- Governance votes.

Wait for **confirmed** (or **finalized** for critical flows) before showing "success" on money paths.

A production UI checklist

- [ ] Every transaction has 5 explicit on-screen states.
- [ ] A **primary + fallback RPC** is wired via env variables.
- [ ] Priority fee mode is visible **before** the signature request.
- [ ] Wallet rejection is handled as a first-class state, not an error.
- [ ] Every confirmation links to **Solana Explorer**.
- [ ] The page is keyboard-operable and color-contrast-safe.
- [ ] iOS MWA is **honestly** disclosed.

HANDS ON

You will harden the Assignment 2 app into a production-grade transfer experience.

Assignment 3 — live recap

- Env-based RPC primary + fallback.
- Graceful handling of rejection, slowness, expiry, and unknown errors.
- Accessible inputs, focus states, ARIA live regions.
- Priority fee helper with Off / Auto / Manual.
- Lighthouse accessibility score **≥ 90**.

Full spec: `assignments/assignment-3.md`.

Quiz 3 — topics covered

- Minimum number of tx states for a production UI.
- Why RPC reliability is a UX concern.
- Priority fee UX best practices.
- When optimistic UI is safe and when it is not.
- Accessibility facts for crypto apps.

> Full quiz with answers: `quizzes/quiz-3.md`.

END OF LECTURE 03

Next: Testing, tooling & Colosseum readiness.

`create-solana-dapp`

`.env`

References

- Production readiness — solana.com/docs/payments/production-readiness
- Confirmation — solana.com/developers/guides/advanced/confirmation
- Priority fees — solana.com/developers/guides/advanced/how-to-use-priority-fees
- WCAG — w3.org/WAI/standards-guidelines/wcag